

Development of real-world driving cycle: Case study of Pune, India

Sanghpriya H. Kamble^a, Tom V. Mathew^{b,*}, G.K. Sharma^a

^a Central Institute of Road Transport, Pune, India

^b Department of Civil Engineering, Indian Institute of Technology Bombay, Powai, Mumbai, Maharashtra 400 076, India

ARTICLE INFO

Keywords:

Driving cycle
Driving pattern
Driving characteristics
Micro-trips
Frequency
Matrices

ABSTRACT

The critical component of all emission models is a driving cycle representing the traffic behaviour. Although Indian driving cycles were developed to test the compliance of Indian vehicles to the relevant emission standards, they neglects higher speed and acceleration and assume all vehicle activities to be similar irrespective of heterogeneity in the traffic mix. Therefore, this study is an attempt to develop an urban driving cycle for estimating vehicular emissions and fuel consumption. The proposed methodology develops the driving cycle using micro-trips extracted from real-world data. The uniqueness of this methodology is that the driving cycle is constructed considering five important parameters of the time-space profile namely, the percentage acceleration, deceleration, idle, cruise, and the average speed. Therefore, this approach is expected to be a better representation of heterogeneous traffic behaviour. The driving cycle for the city of Pune in India is constructed using the proposed methodology and is compared with existing driving cycles.

1. Introduction

Due to rapidly increasing numbers of vehicles and the very limited use of emission control strategies, motor vehicles are emerging as the largest source of urban air pollution globally. Effectiveness of any control strategy depends on accurate emission models. These emission models are often represented by emission factors and are normally expressed in terms of grams of pollutants per unit of distance traveled. The emission factors in turn depend on several other factors such as type of fuel, type of engine, age of the vehicle, driving cycle, etc. The driving cycle is a representative plot of driving behaviour of a given city or a region and is characterized by speed and acceleration. Driving cycle consist of a sequence of several vehicle operating conditions (idle, acceleration, cruise and deceleration) and is considered as a signature of driving characteristics of that city or region. The driving pattern varies from city to city and from region to region. Consequently, the driving cycles developed in a certain region may not be a good representation elsewhere unless the driving characteristics are conspicuously similar.

The Indian driving cycle (IDC) was formulated around late 1985, after extensive road tests by the Automotive Research Association of India (ARAI), Pune. Although some modification were made in the IDC for passenger cars by lowering maximum speed, it still neglects the speed and acceleration greater than 42 km/h and 0.65 m/s² and assumes all vehicle activities to be homogeneous irrespective of variations in traffic and driving characteristics. Therefore, there is a need to study how the driving cycle of a given city or a region varies from the standard cycle from the real-world driving characteristics. Here we develop an urban driving cycle for vehicular emissions and fuel consumption from real-world data.

2. Background

The development of a real-world driving cycle is important for traffic and transport management, vehicular pollution measurement and control, energy and fuel consumption studies (Bata et al., 1994; Andre, 1996). Driving cycles can be categorized as synthesized and actual drive cycles. The former, including the European cycle, Japanese cycle, California seven-mode cycle, and Indian driving cycle are constructed from a number of constant acceleration and constant speed phases. Because of the complex nature of the synthesized cycles, the transitions between various modes are somewhat artificial in nature. These cycles are unsuitable for evaluating fuel consumption due to its gentle acceleration, braking, and long periods spent in stationary mode (Bata et al., 1994).

Actual driving cycles are the cycles derived from the movement of test vehicle on the road under real traffic conditions. These cycles are synthesized from real micro-trips. These driving cycles are more representative of the intended driving pattern and are generally constructed with a simple pattern so as to obtain high repeatability (Bata et al., 1994). Some prominent drive cycles include FTP-75 developed in US (Kruse and Huh, 1973), bus, truck, and composite drive cycles for New York (Bata et al., 1994), and Bangkok-Cycle (Tamasanya et al., 2006). The actual driving cycles are best suitable for heterogeneous traffic conditions that represent sharp acceleration and decelerations, which are the main modes of operation responsible for higher emissions and fuel consumptions. Driving cycles are developed in several parts of the world including America (Esteve-Booth et al., 2002), Europe (Tzirakis et al., 2006), Australia (Kenworthy and Newman 1982), and Asia (JASIC, 1992; Montazeri and Naghizadeh, 2003; Tzeng and Chen, 1998; Ergeneman et al., 1997; Badusha and Ghosh 1999; Nesamani and Subramaniam, 2005; Anand et al., 2005).

The cardinal purpose of a driving cycle is to simulate actual driving characteristics on the road to test the vehicle exhaust emissions and fuel consumption. Therefore, it is unacceptable to analytically synthesize a driving cycle and expecting it to represent the real traffic condition. On the other hand, actual driving cycles can represent the observed driving patterns much better. Although they are hard to run on a chassis dynamometer and have poor repeatability, they are able to provide accurate emission factors. The driving cycles in different countries have varying set of parameters as basis for deriving a drive cycle. This paper is an attempt to propose a methodology to develop driving cycles from the real-world data collected from heterogeneous traffic.

3. Methodology

A robust methodology for developing real-world and city specific driving cycle is proposed. The driving cycle is developed from micro-trips, which are representative of existing traffic conditions. The uniqueness of this methodology is that the driving cycle is constructed considering five important parameters of the time-space profile namely, the percentage acceleration, deceleration, idle, cruise, and the average speed. The steps involved in this methodology are: collection of driving data (speed-time), generation of micro-trips, data analysis, and construction of driving cycle. The methodology is illustrated in the form of flow chart as show in Fig. 1 and the steps are given below.

3.1. Driving data

The speed-time data is required to be collected using chase-car technique. This technique involves random selection of a vehicle in the traffic and survey vehicle simply has to follow the selected vehicle keeping approximately a constant distance during different modes of operation. The chasing procedure should be repeated to obtain large amount of data. The vehicle speed with respect to time should be measured and recorded on all the selected routes. On the selected routes, the data collection could be carried out during peak and off peak period for varying traffic conditions.

3.2. Generation of micro-trips

Development of a drive cycle is based on *micro-trips*. Micro-trip is an excursion between two successive time points at which the vehicle is stopped. This part of motion consists of acceleration, cruise and deceleration modes. By convention, a period of rest is at the beginning and end of a micro-trip. The whole data has to be separated into number of micro-trips. That is the speed-time data collected for a particular stretch having n numbers of segments of different modes of vehicle operation are divided at each idle value.

3.3. Data analysis

The data analysis to be carried out in two parts as: analysis of base data and the analysis of micro-trips. The analysis of base data involves the development of two matrices, namely speed-acceleration frequency matrix MF_b and normalized speed-acceleration matrix MN_b . The speed-acceleration frequency matrix is the frequency of occurrence of acceleration, deceleration, cruise and idle corresponding to a speed values. From MF_b , generate the MN_b matrix by normalizing the entries on percentage basis so that the total of all the velocity-acceleration entries sum to 100. From the normalized MN_b , several parameters representing travel characteristics are computed. These parameters include the percent time in acceleration

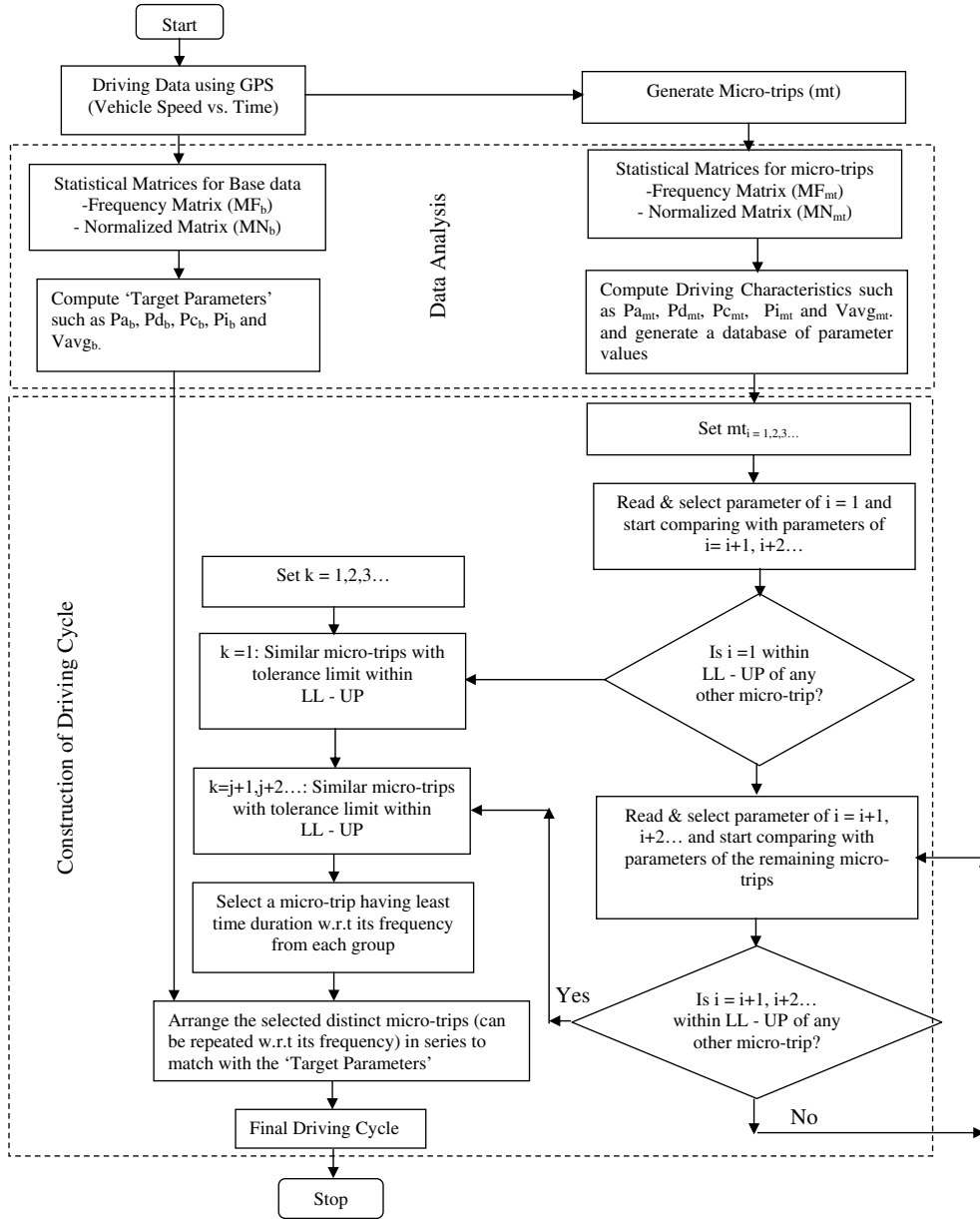


Fig. 1. Methodology for development of the driving cycle.

Pa_b , percent time in deceleration Pd_b , percent time in cruise Pc_b , percent time in idle Pi_b and average velocity $Vavg_b$. These parameters are called as target parameters.

A similar analysis is to be carried out for micro-trips extracted from the base data. Thus, from the micro-trip data the speed–acceleration frequency matrix MF_{mt} and normalized speed–acceleration matrix MN_{mt} are computed as before. Further, parameters representing travel characteristics corresponding to micro-trip data is also computed as before. These parameters from the micro-trips include parameters such as percent time in acceleration Pa_{mt} , percent time in deceleration Pd_{mt} , percent time in cruise Pc_{mt} , percent time in idle Pi_{mt} and average speed $Vavg_{mt}$ from the micro-trip data.

3.4. Construction of driving cycle

To further analyze the data, a computer program is developed, which generates a driving cycle according to the proposed procedure. The program compares all the micro-trips with each other with respect to the calculated parameters of the individual micro-trips within a tolerance limits ranging from a lower limit LL (say about 5%) to an upper limit UL (say about 15%). That is the parameters of the first micro-trip should be compared with parameters of all other micro-trips. Likewise the

parameters of the second micro-trip should be compared with the parameters of all other micro-trips. This procedure has to be repeated with all other remaining micro-trips. The comparison of the parameters are required, in order to reduce the number of micro-trips from the original size K to a very smaller size k which still can represent the entire data and sufficient to obtain the feasible drive cycle duration. This comparison starts from the lower limit of the tolerance in order to check the amount of reduction in number of micro-trips. If after comparison, the number of micro-trips obtained is high, then the tolerance limit should be increased.

Consider a set of micro-trips, say $i = 1, 2, 3, \dots$. The micro-trips, which are similar, to be considered under one group. Group 1 ($g_i = 1$) micro-trips have similar parameters value but differ from other group. If one micro-trip appears to be similar to say five other micro-trips from the total number of micro-trips, its frequency is said to be five, likewise with other groups the procedure is repeated. From each group the computer program selects a micro-trip that is distinct in nature and has least time duration. Out of K original micro-trips, sub set of size k micro-trips are selected with their respective frequencies. Then a real-world driving cycle is build by selecting multiples of micro-trips from k with their respective frequencies of occurrences to match the target parameters, namely Pa_b , Pd_b , Pc_b , Pi_b , and $Vavg_b$ of the base data. In order to match the parameters of the base data for the development of final driving cycle, all the selected k distinct micro-trips are first taken into consideration and are connected in series. If the overall parameters of these k micro-trips do not match with the target parameters, then according to the frequency of each micro-trip available, the micro-trips are repeated. Once the parameters match, the final real-world driving cycle is obtained by concatenating the individual micro-trips. The total duration of the driving cycle should be long enough to describe all traffic situations and obtain the emissions sufficiently.

4. Case study

This methodology is illustrated using a case study based on the data collected from Pune city, in India. Pune city is an important urban center in Maharashtra and a rapidly growing metropolis of the country with highest two-wheelers. With introduction of thousands of vehicles per month, the traffic congestion in the city is increasing alarmingly. As a consequence, average speeds on the city roads are greatly impaired and range between 15 km/h and 35 km/h. The traffic varies considerably within a short distance, so in order to understand the traffic variations and its impact; the method is illustrated in details by analyzing the data in the ratio of 2:1.

4.1. Driving cycle development

Extensive, time and speed data from five major roads measuring approximately about 55 km from Pune city was collected using GPS (Table 1). From the base data of Pune city, the speed–acceleration frequency matrix (MF_b) was generated and is given in Table 2. From this, the normalized speed–acceleration matrix from the base data (MN_b) and is given in Table 3.

The target parameters are calculated from the MN_b . The target parameters obtained are percentage acceleration (14.58), percentage de-acceleration (12.23), percentage cruise (54.63), and percentage idle (18.61). The average velocity is calculated (21.05 kmph) directly from the speed data.

Similarly, the analysis is carried out for the micro-trips. The base data is separated into micro-trips. The number of micro-trips obtained are $K = 46$. Details of some of the micro-trips are shown in the Fig. 2. These micro-trips are part of speed–time profile, which starts and ends at zero values representing real driving pattern. The speed–acceleration frequency matrix (MF_{mt}) and speed–acceleration normalized matrix (MN_{mt}) are generated for all the individual micro-trips. For example, the parameters calculated from 9th micro-trip are the percentage acceleration of 16, percentage de-acceleration of 13, percentage cruise of 70, percentage idle of 1 and an average velocity of 5 kmph. This procedure is repeated for all other micro-trips. With the help of specially designed computer software, the selected parameters of all the micro-trips such as Pa_{mt} , Pd_{mt} , Pc_{mt} , Pi_{mt} , and $Vavg_{mt}$ are compared with each other with respect to upper and lower limit (UL-LL). After comparison, out of $K = 46$ micro-trips, $k = 17$ micro-trips are selected with their respective frequencies. The selected distinct micro-trips with their parameters are given in Table 4. In this table frequency indicates the number of times the selected distinct micro-trip similar to the micro-trips in their respective groups. The details of all the selected distinct micro-trips having least time duration are shown in Fig. 3.

Table 1
Selected routes that represents real traffic conditions.

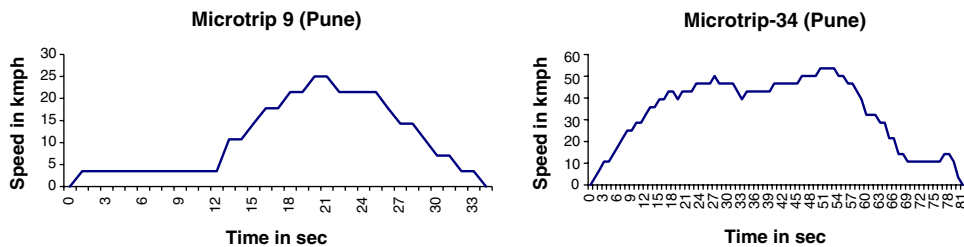
Sl. no.	From	To	Distance (km)
1	CIRT	Warje	19.5
2	Warje	CIRT	20.8
3	Swargate	Shivajiagar	4.3
4	Corporation	Pune Stn	3.5
5	Hadapsar	Bhariobanala	6.1
Total			55.0

Table 2The speed–acceleration frequency matrix (MF_b) from base data.

Speed (kmph)	Acceleration (km/h/s)																				Total	
	−10	−9	−8	−7	−6	−5	−4	−3	−2	−1	0	1	2	3	4	5	6	7	8	9		10
0–5				1			6	26			1286			65	17			1				1402
5–10				7			7	13			198				48		1					274
10–15	6			17	8		18	18			480			98	29		2	2				678
15–20		5		10	3		5	20			263			2	70		1				1	380
20–25		9		17	6		45	25			559			47	89		3	1				801
25–30		5		10	7		37	2			278			38	22		8	13			2	422
30–35		2		4	3		22	39			230			23	20		2	1			4	350
35–40		4		8	8		82	30			366			17	99		1				1	616
40–45		1			3		31	17			180			0	31		2	1			1	267
45–50				1	1		29				130			14	20						2	197
50–55		3			2		32	20			212			5	23		2					299
55–60					1		13				29				9							52
60–65		1					11				12				2							26
65–70							2				7				1							10
70–75							1															1
Total	36	0	0	75	42	0	341	210	0	0	4230	0	0	309	480	0	22	19	0	0	11	5775

Table 3The normalized speed–acceleration matrix (MN_b) from the base data.

Speed (kmph)	Acceleration (km/h/s)																				Total	
	−10	−9	−8	−7	−6	−5	−4	−3	−2	−1	0	1	2	3	4	5	6	7	8	9		10
0–5				0.17			0.14	0.45			18.61			1.13	0.29			0.17				21
5–10				0.12			0.12	0.23			3.43				0.83		0.17					5
10–15	0.14			0.29	0.14		0.31	0.31			8.31			1.70	0.52		0.35	0.35				12
15–20	0.87			0.17	0.52		0.87	0.35			4.55			0.35	1.21		0.17			0.17		9
20–25	0.16			0.29	0.14		0.78	0.43			9.68			0.81	1.54		0.52	0.17				15
25–30	0.87			0.17	0.12		0.65	0.35			4.81			0.66	0.39		0.14	0.23			0.35	9
30–35	0.35			0.69	0.52		0.39	0.68			3.98			0.40	0.35		0.35	0.17			0.69	9
35–40	0.69			0.14	0.14		1.42	0.52			6.34			0.29	1.71		0.17				0.17	12
40–45	0.17				0.52		0.54	0.29			3.12				0.54		0.35	0.17			0.17	6
45–50				0.17	0.17		0.52				2.68			0.24	0.35						0.35	4
50–55	0.52				0.35		0.55	0.35			4.54			0.87	0.40		0.35					8
55–60					0.17		0.23				1.37				0.16							2
60–65	0.17						0.19				0.83				0.35							2
65–70							0.35				0.99				0.17							2
70–75							0.17															0
Total	0.62	0	0	1.30	0.73	0	5.95	3.64	0	0	73.25	0	0	5.36	8.31	0	0.39	0.33	0	0	0.19	114

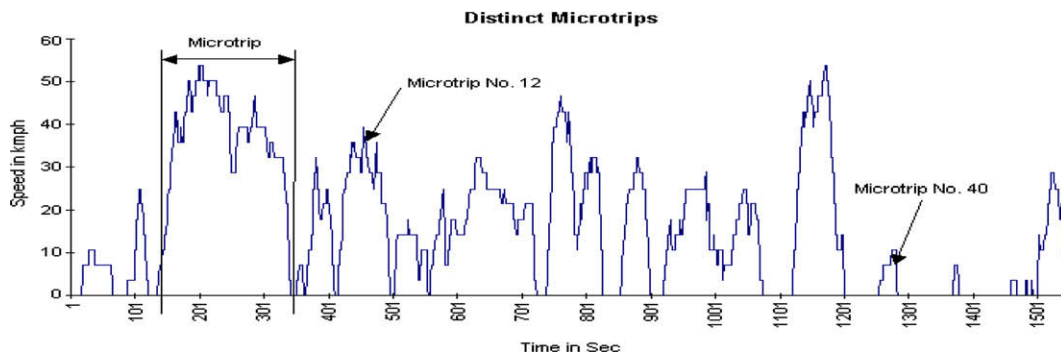
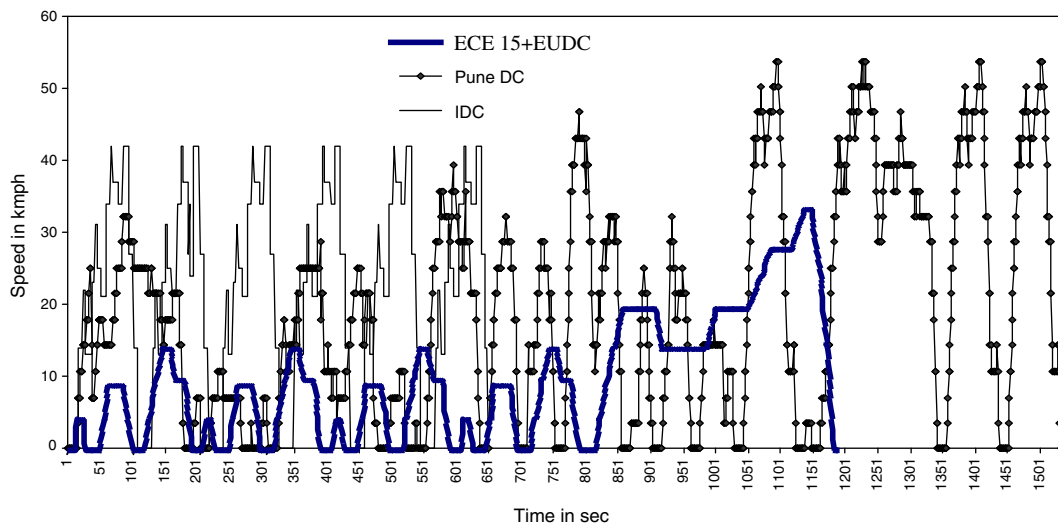
**Fig. 2.** Typical micro-trips obtained from the actual data.

The real-world driving cycle is build by selecting multiples of k micro-trips with their respective frequencies of occurrences to match the parameters of the base data. After the selection of k micro-trips, using the computer program micro-trips are randomly selected with respect to their frequencies and arranged in series to obtain the real word driving cycle whose parameters almost match the target parameters. The total duration of the real-world driving cycle achieved is 1533 s. The parameters values obtained are percent acceleration of 14.18, percent deceleration of 11.48, percent cruise of 56.25, percent idle of 18.09 and average velocity of 19.55 kmph. The real word driving cycle thus developed is as shown in the Fig. 4 along

Table 4

Selected distinct micro-trips and their parameter values.

Group no.	Micro-trip no.	Avg_vel (kmph)	% Acceleration	% Deceleration	% Cruise	% Idle	Time (s)	Frequency
1	1	7	4	4	88	4	49	1
2	2	11	15	18	62	6	34	1
3	34	34	23	19	56	2	81	4
4	42	3	0	0	50	50	4	7
5	41	4	0	0	80	20	10	2
6	20	21	17	11	67	4	46	7
7	9	37	16	13	70	1	208	2
8	29	16	13	8	77	1	156	2
9	11	5	8	0	75	17	12	1
10	12	18	18	20	58	4	45	2
11	13	26	17	16	65	2	82	5
12	14	12	11	6	79	4	53	2
13	15	20	10	7	82	1	165	1
14	19	28	26	20	51	2	88	2
15	45	18	14	14	67	5	43	4
16	36	7	7	3	83	7	29	1
17	40	5	10	10	60	20	10	1

**Fig. 3.** Distinct micro-trips separated from the from the actual data.**Fig. 4.** Comparison of Pune cycle with IDC and European cycle.

with the standard IDC and the European driving cycle (ECE-15 + EUDC). The driving cycle for Pune is very different from the standard driving cycle characterized by higher speed and steep acceleration-deceleration.

4.2. Evaluation of the driving cycle

To evaluate the developed Pune driving cycle, comparison with the existing standard driving cycles namely European and IDC has been done. The comparison is based on key parameters as illustrated in Table 5. It can be observed that, the average speed in Pune driving cycle was 43% and 16% lower than average speed in ECE-15 + EUDC cycle and IDC, whereas average acceleration was higher of the order of 587% and 120%. The average deceleration was 479% and 133% higher than average deceleration in ECE-15 + EUDC cycle and IDC. Although the time spent in acceleration for Pune city was about 22% and 64% lower than in ECE-15 + EUDC cycle and IDC, the magnitudes of acceleration are quite sharp in Pune cycle. Also true for decelerations.

Since parameters such as acceleration and deceleration have greater impact on emissions and fuel consumption, a comparison of the important statistics of ECE-15 + EUDC cycle and IDC is made with that of the Pune cycle and is tabulated in Table 6 for acceleration and Table 7 for deceleration. It is observed that in ECE-15 + EUDC, acceleration was always less than 1 m/s^2 (avg. acceleration 0.541), where as in IDC and Pune cycle (average acceleration is 1.69 and 3.72 m/s^2) it was found to be 73% of the time was in acceleration less than 1 m/s^2 , where as in rest of the 27% of time was in acceleration greater than 1 m/s^2 .

From Table 7, in ECE-15 + EUDC cycle with an average deceleration of -0.789 m/s^2 the deceleration of less than 1 m/s^2 occurs 94% of the time and only 6% of the time the deceleration was greater than 1 m/s^2 . In Pune cycle with average deceleration of -4.57 m/s^2 , for 75% of the time, deceleration was less than 1 m/s^2 , where as rest 25% of time the deceleration was greater than 1 m/s^2 .

Table 8 compares percentage time spent in different speed range by the ECE-15 + EUDC cycle, IDC and Pune Cycle. It is seen that driving with speeds $< 10 \text{ km/h}$ occurs 32% of time in Pune cycle, whereas in ECE-15 + EUDC, this is not observed. The time spent between 10 and 20 km/h is 22% which is 3 times more compared to ECE-15 + EUDC cycle. Due to heterogeneous traffic leading to congestions and unsustainable transport system, the time spent in speed $\leq 30 \text{ kmph}$ is more making trips short with low speeds.

From an overall consideration, it can be concluded that the traffic in India has large fluctuations due to heterogeneity and congestion leading to higher variation in speed, acceleration, and deceleration characteristics. Therefore, high emissions and

Table 5

Comparison of parameters from Pune, IDC, and European driving cycle.

Driving cycles	V_{avg} (km/h)	% Idle	% Cruise	% Acceleration	% Deceleration
ECE-15 + EUDC	33.4	23.7	42.2	18.3	15.8
IDC	21.9	16.52	10.43	38.89	34.26
Pune	19.55	18.09	56.25	14.18	11.48

Table 6

Comparison of acceleration parameters from Pune, IDC, and European driving cycle.

Driving cycles	Acc_{avg} (m/s^2)	% Time in acceleration	% Time in acceleration $< 1 \text{ m/s}^2$	% Time in acceleration $> 1 \text{ m/s}^2$
ECE-15 + EUDC	0.541	18.3	18.3	0
IDC	1.69	38.89	28.01	10.88
Pune	3.72	14.18	10.29	3.88

Table 7

Comparison of deceleration parameters from Pune, IDC, and European driving cycle.

Driving cycles	Dec_{avg} (m/s^2)	% Time in deceleration	% Time in deceleration $< -1 \text{ m/s}^2$	% Time in deceleration $> -1 \text{ m/s}^2$
ECE-15 + EUDC	-0.789	15.8	14.83	0.97
IDC	-1.96	34.26	26.54	7.72
Pune	-4.57	11.48	8.59	2.63

Table 8

Comparison of percentage time spend in different velocity range from Pune, IDC, and European driving cycle.

Driving cycles	$0 \leq V \leq 10$	$10 \leq V \leq 20$	$20 \leq V \leq 30$	$30 \leq V \leq 40$	$V \geq 40$
ECE-15 + EUDC	0	7.2	20.7	31.6	40.5
IDC	10	20	30	30	10
Pune	32.44	22.00	21.34	12.01	12.27

Table 9

Comparison of parameters from driving cycles developed from various data sets.

City	Data	Cycle duration (s)	Average velocity (kmph)	Vel _{max} (kmph)	Acc _{mean} (m/s ²)	Acc _{max} (m/s ²)	Dec _{mean} (m/s ²)	Dec _{max} (m/s ²)
Pune	First 2/3	1274	20.51	35.11	3.73	7.22	−4.51	−10.92
	Second 1/3	1242	19.61	53.69	3.66	7.04	−4.76	−10.92
	Third complete set	1533	19.55	53.69	3.73	14.26	−4.58	−10.92

Table 10

Comparison of parameters from the driving cycles developed from various data sets and the corresponding base data.

Parameters	First 2/3 data		Second 1/3 data		Complete data	
	Base data	Driving cycle	Base data	Driving cycle	Base data	Driving cycle
P_a	15.04	13.46	14.18	16.65	14.78	14.18
P_d	12.35	11.1	12.19	12.67	12.3	11.48
P_c	56.66	59.31	49.73	47.83	54.41	56.25
P_i	15.99	16.12	24.01	22.86	18.55	18.09

Table 11

Comparison of percent time spent in different speed ranges.

Driving cycles	0 < V < 10	10 < V < 20	20 < V < 30	30 < V < 40	>40
First 2/3	31.63	20.80	20.57	13.74	13.27
Second 1/3	59.82	11.43	9.50	5.72	13.53
Third complete set	32.44	22.00	21.34	12.01	12.27

fuel consumption can be expected from the emission factors developed from this driving cycle, which could be more realistic.

4.3. Validation of the driving cycle

To validate the developed Pune driving cycle, the data was divided into two parts: the first 2/3 data and the last 1/3 data. Three separate driving cycles are developed from the first set, second set, and the complete set of data. From these driving cycles, the parameters obtained are summarized in Table 9. It can be observed from the table that parameters are closely matching in all the cases indicating the sample size is sufficient and that the methodology is reproducible.

The real-world driving cycle parameters are also compared with parameters of base data in order to check whether the driving cycle parameters closely match with the parameters of the base data. Table 10 gives the details of the driving parameters obtained from the real-world driving cycles and the base data. The base data parameters may be considered as the target parameters to which the driving cycle parameter should conform. It can be observed that the values of the parameters of base data and driving cycles almost match with each other indicating that the driving cycle truly represent driving characteristics of the city of Pune.

Finally, the percent time spent in different speed ranges gives an idea about the traffic conditions in the city and the same is tabulated in Table 11. It is seen that the time spent in driving with speeds less than 10 km/h is the highest (59%). Though, the time spent in speed between 20 kmph and 40 kmph is 6% to 20%, still there exist a problem of traffic congestion because 80% of the time spent is below 20 kmph. Therefore, larger values in both emissions and fuel consumption can be expected from the vehicular traffic.

5. Conclusion

A methodology for development of driving cycle using micro-trips extracted from real-world driving data is developed in this paper. The uniqueness of the methodology is that the driving cycle is constructed considering important parameters of the time–space profile; the percentage acceleration, deceleration, idle, cruise, and the average speed. Therefore, this approach is expected to be a better representation of heterogeneous traffic behaviour. The driving cycle for the city of Pune in India is constructed using the proposed methodology and is validated and compared with the existing driving cycles. The proposed driving cycle exhibits distinctly different pattern when compared with the standard cycles such as ECE-15 + EUDC. These cycles, therefore, cannot be a true representation of the actual traffic condition in the city of Pune. Consequently, it is not reasonable to expect the emission factors built on these cycles give accurate emissions. The observations are likely to be true for any given city and therefore there is a need to understand how the driving cycle differs from city to city and for various mode of traffic.

References

- Anand, V.K., Rekha, R.S., Chirag, P.S., 2005. GPS based methodology for drive cycle determination. SAE Transactions 114, 233–239.
- Andre, M., 1996. Driving cycle development: characterization of the methods. In: SAE International Spring Fuels and Lubricants Meeting. SAE Paper 961112, Washington, DC.
- Badusha, A.A., Ghosh, B., 1999. Driving Cycle for Buses in Delhi City, Society of Automobile Engineers. SAE Technical Paper Series No. 990036, Washington, DC.
- Bata, R., Yasser, Y., Wang, W., Lyons, D., Gambino, M., Rideout, G., 1994. Heavy Duty Testing Cycles: Survey and Comparison. SAE Technical Paper Series No. 942263, Washington, DC.
- Ergeneman, M., Sorusbay, C., Goktan, A., 1997. Development of a driving cycle for the prediction of pollutant emissions and fuel consumption. International Journal of Vehicle Design 18, 391–399.
- Esteve-Booth, A., Muneer, T., Kubie, J., Kirby, H., 2002. A review of vehicular emission models and driving cycles. Proceedings of the Institution of Mechanical Engineers: Part C: Journal of Mechanical Engineering Science 216, 777–797.
- JASIC, 1992. Automobile Type Approval Handbook for Japanese Certification. Japan Automobile Standards Internationalization Center, Tokyo.
- Kenworthy, J., Newman, P., 1982. A driving cycle for Perth: methodology and preliminary results. In: Proceedings of the Joint SAE-A/ARRB 2nd Conference on Traffic, Energy and Emissions, Melbourne.
- Kruse, R.E., Huh, T.A., 1973. Development for the federal urban driving cycle. US Environmental Protection Agency, SAE Paper 730553, Washington, DC.
- Montazeri, Gh.N., Naghizadeh, M., 2003. Development of Car Driving Cycle for Simulation of Emissions and Fuel Economy, ESS2003, Delft.
- Nesamani, K.S., Subramaniam, K.P., 2005. Impact of Real World Driving Characteristics on Vehicular Emissions Institute of Transportation Studies, University of Irvine.
- Tamasanya, S., Chunpaibulpattana, S., Attajariyakul, S., 2006. Development of automobile Bangkok driving cycle for emissions and fuel consumption assessment. In: Second Joint International Conference on Sustainable Energy and Environment, Bangkok.
- Tzirakis, E., Pitsas, K., Zannikos, F., Stournas, S., 2006. Vehicle emissions and driving cycles: comparison of the Athens driving cycle (Adc) with Ece-15 and European driving cycle (EDC). Global NEST Journal 8, 282–290.
- Tzeng, G.H., Chen, J.J., 1998. Development of Taipie motor cycle driving cycle for emission and fuel economy. Transportation Research Part D 3, 19–27.